Grade Test: Quiz 1

Test Information

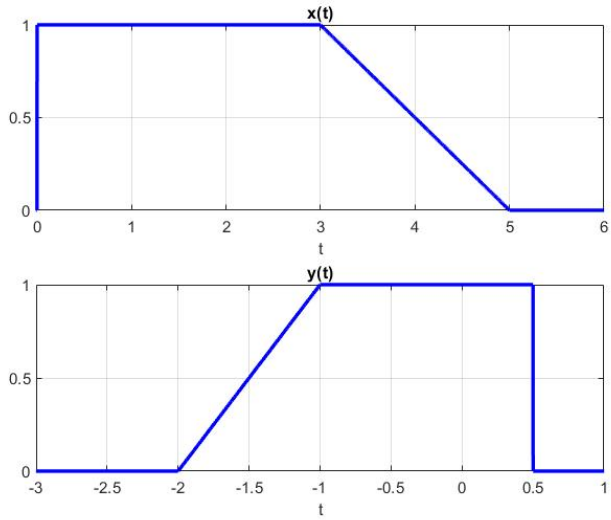
Instructions

- The duration of the quiz is 25 minutes.
- Once you submit an answer, you cannot scroll back to previous questions.

QUESTION 1: MULTIPLE CHOICE

10 out of 10 points

Consider the signals x(t) and y(t) shown in the figure below. How are these signals related?



Given Answer: y(t)=x(-2t+1)

Correct Answer: y(t)=x(-2t+1)

QUESTION 2: CALCULATED FORMULA | **5** **out of 5 points**

Find the power of the signal 5sin(191t+2).

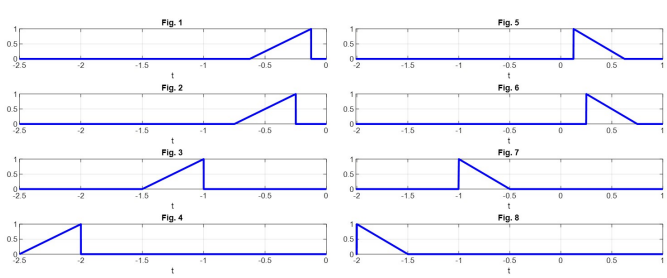
Given Answer: 12.5

Correct Answer: 12.5

QUESTION 3: MULTIPLE CHOICE | **10** **out of 10 points**

Consider the signal $x(t) = \mathrm{tri}\left(\frac{t}{4}\right) \times \mathrm{u}(-t)$.

Which one of the below figures corresponds to the signal $x(-8t+2)$?



Given Answer: Fig. 6

Correct Answer: Fig. 6

QUESTION 4: MULTIPLE CHOICE 10 **out of 10 points**

Consider the signal $x(t) = \text{rect}\left(\frac{t}{8}\right) \times \text{rect}\left(\frac{t-1}{3}\right)$.

x(t) can be written as:

Given Answer: $\text{rect}\left(\frac{t-1}{3}\right)$

Correct Answer: $\text{rect}\left(\frac{t-1}{3}\right)$

QUESTION 5: CALCULATED FORMULA 10 **out of 10 points**

What is the energy of the signal $a\text{rect}\left(\frac{t-b}{c}\right)$ where a=5, b=0 and c=9?

Given Answer: 225

Correct Answer: 225

QUESTION 6: TRUE/FALSE 5 **out of 5 points**

The system $y(-t) = x(t)$ is causal.

Given Answer: False

Correct Answer: False

QUESTION 7: TRUE/FALSE 5 **out of 5 points**

The system $y(t) = x(t^2)$ is time invariant.

Given Answer: False

Correct Answer: False

QUESTION 8: CALCULATED FORMULA | 10 | **out of 10 points**

Consider the signal: $x(t) = \text{rect}\left(\frac{t}{2a}\right) + 3\text{rect}\left(\frac{t}{a} - \frac{1}{2}\right)$.

What is the duration of the signal $-x\left(\frac{-t}{b} + 7\right)$ for a=3 and b=8?

Given Answer: 48

Correct Answer: 48

QUESTION 9: MULTIPLE CHOICE | 10 | **out of 10 points**

Consider the signal x(t)=u(2t).

How can this signal be classified?

Given Answer: Power Signal.

Correct Answer: Power Signal.

QUESTION 10: CALCULATED FORMULA | 10 | **out of 10 points**

The signal x(t) extends between -0.25 sec and +0.25 sec. The signal has an energy of 2 Joules.

What is the energy of the signal $x(t) + x(t+a) + bx(t-a)$ where a=4 and b=2?

Given Answer: 12

Correct Answer: 12

QUESTION 11: TRUE/FALSE | 5 | **out of 5 points**

The system $y(t) = e^{t}x(|t|)$ is linear.

Given Answer: True

Correct Answer: True

QUESTION 12: MULTIPLE CHOICE

10 out of 10 points

Consider the signal x(t)=u(t+12)-u(t+8).

Write down x(t) in terms of the rectangular function.

Given Answer: $\text{rect}\left(\frac{t+10}{4}\right)$

Correct Answer: $\text{rect}\left(\frac{t+10}{4}\right)$

FEEDBACK AND NOTES FOR ATTEMPT

Feedback to Learner

Paragraph Arial 3 (12pt)

Mashups

Path: p Words:0

Grading Notes

Paragraph Arial 3 (12pt)

Mashups

Path: p Words:0